CONTINUOUS REPRESENTATION BEFORE RECONSTRUCTION: AN ANALYTICAL FRAMEWORK FOR 3D PARALLEL MRI SIMULATION

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ABSTRACT

Simulated k-space serves as ground truth for evaluating, and increasingly training, MRI reconstruction methods, yet most simulators still rely on a voxel grid. The spectrum of a sampled object contains periodic replicas that overlap the acquired band, and simulating motion requires resampling the volume at every pose. Existing analytical phantoms offer alternatives to grid-based simulation but are less flexible and efficient in representing spatially varying anatomical intensities. We introduce a simulator framework which represents the subject through continuous representation with a mixture of anisotropic Gaussians fitted to an anatomical volume[1]. Fourier encoding, rigid motion, coil sensitivities and sequence contrast then act on the Gaussian parameters in closed form, giving multi-coil k-space at arbitrary sampling locations and poses without a grid. On a 3D Shepp–Logan phantom with varying intensity, our framework provides higher anatomy fidelity with 13.7 dB higher PSNR than the polyhedral phantom, and 3.2 dB higher than a tetrahedral phantom with four times as many parameters, while using 40 times less GPU memory and evaluating k-space over 400 times faster. On BrainWeb anatomy, voxel-based simulation exhibits resolution-dependent accuracy ceilings that persist as noise decreases while image-based motion simulation adds resampling errors whereas analytical encoding avoids these simulation-induced errors even under rigid motion.

***Index Terms*—** MRI simulation, analytical phantom, Gaussian primitives, parallel imaging, motion

1. INTRODUCTION

MRI reconstruction is increasingly moving toward continuous representations, from implicit neural representations [1] to Gaussian models [2, 3]. Yet the k-space data used to train and evaluate these methods are often simulated from voxelized anatomy, creating a mismatch between continuous reconstruction models and their discretized references. Simulators such as JEMRIS, MRiLab, KomaMRI and SNAKE [4, 5, 6, 7] typically rely on voxel grids. Fourier encoding of a sampled object introduces spectral replicas that can overlap the acquired band (Fig. 1b). Further, gridded image-based motion simulation results in interpolation errors. Benchmarks evaluated on such data may partly reflect simulator artifacts, thereby giving over-optimistic performance of the reconstruction method being evaluated (Fig. 1c and Fig. 4), a concern related to the *inverse crime* [8].

Analytical phantoms address this limitation by evaluating Fourier signals directly from continuous objects. Closed-form models exist for ellipsoids [9], polygons and Bézier curves [8], and polyhedra [10], while BART digital reference objects incorporate analytical coils and signal models [11]. However, these geometric phantoms typically rely on piecewise-constant regions, requiring further subdivision or richer basis functions to represent spatially varying intensities. For polyhedral models, increasing anatomical detail also increases the number of facets involved in k-space evaluation, making finely detailed simulations computationally expensive.

We propose an analytical MRI simulation framework based on anisotropic Gaussian mixtures fitted to anatomical volumes. Gaussians admit closed-form Fourier transforms, preserve their form under rigid motion, and allow Fourier-basis coil modulation to be expressed as shifts in frequency. Sequence-dependent contrast is incorporated through tissue-specific coefficients, reusing the same fitted geometry. Together, these properties enable efficient multi-coil k-space simulation at arbitrary sampling locations and rigid poses without voxelization or image resampling. Our contributions are: (i) an efficient analytical simulation framework combining anatomical detail, sequence contrast, multi-coil encoding and rigid motion; (ii)

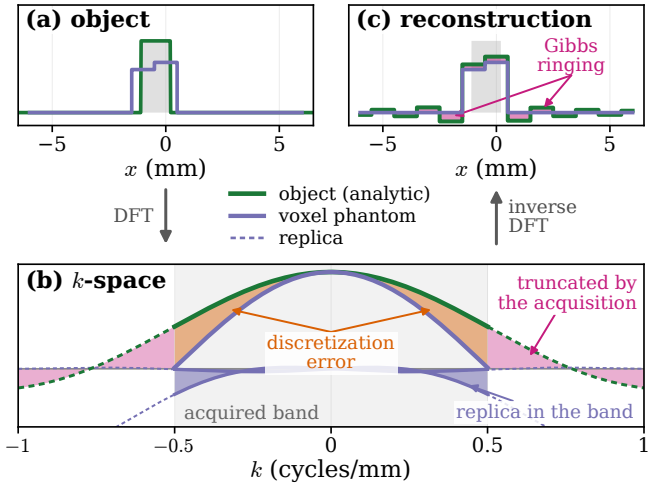


Fig. 1. Grid artifacts for a 1.3 mm sharp structure on a 1 mm grid. (a) Continuous object and its voxel phantom (1 mm voxel averages). (b) Real part of their spectra. Inside the acquired band, the voxel phantom deviates from the object (discretization error: box attenuation and spectral replicas); beyond the band, the acquisition truncates the object spectrum. (c) Inverse DFT of fully sampled data: analytical data exhibit Gibbs ringing, as in a real acquisition, whereas the DFT of the voxel phantom returns the phantom without ringing.

[1]Code and reproduction scripts: `https://github.com/JoosenLi/AnalyticalGaussMRI`.

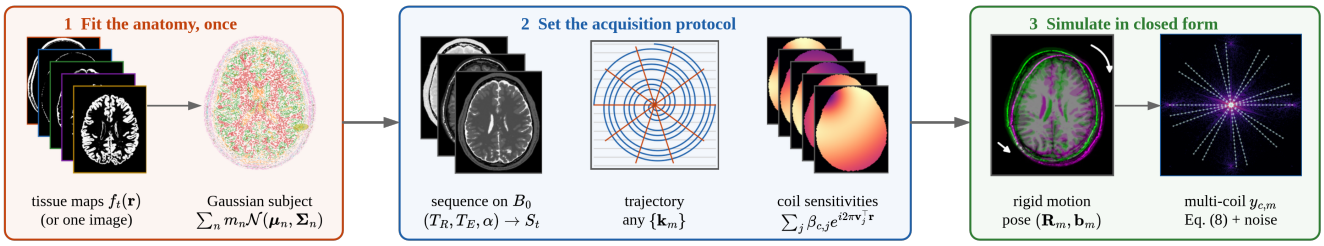


Fig. 2. The simulator in three steps. (1) A Gaussian subject is fitted once to tissue maps (or to a single contrast). (2) The protocol is set: field strength and sequence fix each tissue's signal, the trajectory any set of sampling locations, and the receive coils a Fourier expansion of any sensitivity maps. (3) The multi-coil signal is evaluated in closed form, with a rigid pose per shot or per sample acting on the query frequency.

a quantitative benchmark against polyhedral and tetrahedral phantoms, demonstrating accurate anatomical representation with substantially lower computational cost; and (iii) a theoretical and experimental analysis of discretization and motion-resampling errors, including their impact on reconstruction accuracy.

2. ANALYTICAL MRI SIMULATION

We outline our framework in Fig. 2, where the anatomy is fitted once, the acquisition protocol is set, and the signal is evaluated in closed form.

2.1. Anatomy representation and fitting

Positions $\mathbf{r}$ are in mm and frequencies $\mathbf{k}$ in cycles/mm, with $\hat{\rho}(\mathbf{k}) = \int \rho(\mathbf{r}) e^{-i2\pi \mathbf{k}^\top \mathbf{r}}\, d\mathbf{r}$. In this work, we model the sequence-dependent MR signal as a sum of tissue maps $f_t(\mathbf{r})$, each weighted by its contrast $S(\theta, \eta_t)$, which depends on the sequence parameters $\theta = (T_R, T_E, \alpha)$ and the tissue parameters $\eta_t = (\mathrm{PD}_t, T^*_{2,t}, T_{1,t})$:

$$\rho_\theta(\mathbf{r}) = \sum_t S(\theta, \eta_t)\, f_t(\mathbf{r}). \tag{1}$$

For a spoiled gradient-recalled echo (GRE) sequence in steady state,

$$S(\theta, \eta_t) = \mathrm{PD}_t e^{-T_E/T^*_{2,t}} \frac{\sin\alpha\,(1 - E_t)}{1 - E_t \cos\alpha}, \tag{2}$$

with $E_t = e^{-T_R/T_{1,t}}$. Each tissue map is represented by a mixture of Gaussian primitives [12, 2],

$$f_t(\mathbf{r}) \approx \sum_{n \in \mathcal{I}_t} m_n \mathcal{N}(\mathbf{r}; \boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \quad \boldsymbol{\Sigma}_n = \mathbf{U}_n\, \mathrm{diag}(\sigma_n^2) \mathbf{U}_n^\top, \tag{3}$$

where $\boldsymbol{\mu}_n$ is the centre, σ_n the positive standard deviations, and $\mathbf{U}_n$ a rotation matrix parameterized by a unit quaternion. The Gaussian $\mathcal{N}$ has unit integral, so m_n specifies the integrated weight of each primitive, independent of its shape.

Each mixture is fitted to a voxelized tissue map (e.g., BrainWeb [13]) with Adam through a differentiable tile-binned renderer, minimizing the squared image-domain residual regularized by a smoothed total-variation penalty $\mathcal{R}_t = \lambda\, \mathbb{E}_{\mathbf{r} \sim \mathcal{U}(\Omega)} \sqrt{\mathrm{d}r^2 \|\nabla \tilde{f}_t(\mathbf{r})\|^2 + \delta^2}$, where λ is the regularization weight, $\tilde{f}_t$ the fitted mixture, $\mathrm{d}r$ the voxel size of the source volume and $\delta = 10^{-3}$ a smoothing constant. The expectation is estimated at points drawn uniformly from Ω, the field of view of the source volume, using the analytical gradient of the mixture.

The Fourier transform of the Gaussian mixture is available in closed form:

$$\hat{\rho}_\theta(\mathbf{k}) = \sum_n S(\theta, \eta_{\tau(n)})\, m_n\, e^{-2\pi^2 \mathbf{k}^\top \boldsymbol{\Sigma}_n \mathbf{k}}\, e^{-i2\pi \mathbf{k}^\top \boldsymbol{\mu}_n}, \tag{4}$$

where $\tau(n)$ identifies the tissue associated with primitive n. Fitting the tissue maps separately allows different sequence contrasts to reuse the same set of primitives by updating only $S(\theta, \eta_{\tau(n)})$. Alternatively, a single-contrast image can be fitted directly, in which case m_n directly represents the signal intensity.

2.2. Analytical encoding with coil sensitivities and motion

Receive coil sensitivities $s_c(\mathbf{r})$ for coil c are approximated using a shared compact support of Fourier basis across channels [8] given by $\sum_j \beta_{c,j} e^{i2\pi \mathbf{v}_j^\top \mathbf{r}}$. The coefficients $\beta_{c,j}$ are fitted once by regularized least squares to measured or simulated sensitivity maps over the head support. By the modulation theorem, the signal of coil c at sample $\mathbf{k}_m$ is

$$y_{c,m} = \sum_j \beta_{c,j} \hat{\rho}_\theta(\mathbf{k}_m - \mathbf{v}_j). \tag{5}$$

Motion is modeled as $\mathbf{r} \mapsto \mathbf{A}_m \mathbf{r} + \mathbf{b}_m$, where $\mathbf{A}_m$ is a rotation matrix (rigid-body motion) and $\mathbf{b}_m$ a global translation. Therefore, in the acquisition frame of reference:

$$\rho_m(\mathbf{r}) = |\det \mathbf{A}_m|^{-1} \rho_\theta\big(\mathbf{A}_m^{-1}(\mathbf{r} - \mathbf{b}_m)\big) \tag{6}$$

$$\hat{\rho}_m(\mathbf{k}) = e^{-i2\pi \mathbf{k}^\top \mathbf{b}_m} \hat{\rho}_\theta(\mathbf{A}_m^\top \mathbf{k}) \tag{7}$$

The complete forward model is

$$y_{c,m} = \sum_j \beta_{c,j} e^{-i2\pi (\mathbf{k}_m - \mathbf{v}_j)^\top \mathbf{b}_m} \hat{\rho}_\theta(\mathbf{A}_m^\top (\mathbf{k}_m - \mathbf{v}_j)) + n_{c,m}. \tag{8}$$

The coils stay in the scanner frame, so the pose acts on the shifted frequency $\mathbf{k}_m - \mathbf{v}_j$. The noise is complex Gaussian, $(n_{1,m}, \ldots, n_{C,m})^\top \sim \mathcal{CN}\big(\mathbf{0}, (\bar{E}/\mathrm{SNR})\, \boldsymbol{\Psi}\big)$, where $\bar{E}$ is the mean noiseless signal power across samples and coils, and $\boldsymbol{\Psi}$ is the coil noise covariance normalized to unit mean diagonal, so that SNR is a power ratio [7].

3. EXPERIMENTS AND RESULTS

3.1. Anatomical representation performance

We compare Gaussian mixtures with the analytical polyhedral model [10], and two tetrahedral variants using a 3D Shepp–Logan

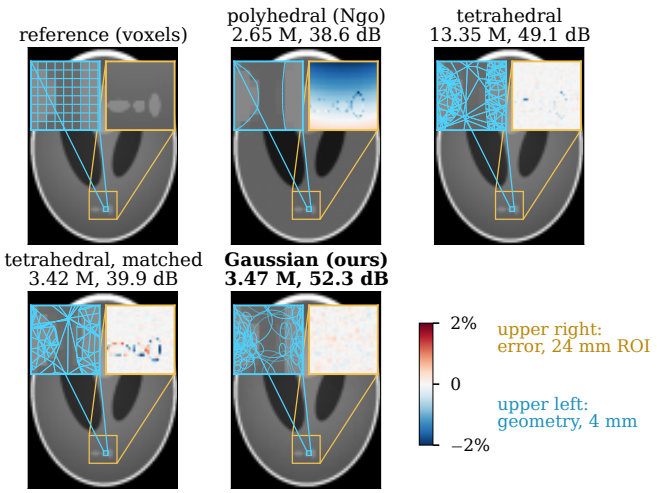


Fig. 3. Representation of the smoothly varying 3D Shepp–Logan phantom (axial slice). Model parameter counts and whole-volume PSNRs are shown above each image. Upper-right insets: the residuals within a 24 mm ROI ($\pm 2\%$ of the reference maximum). Upper-left insets: the underlying geometry within a 4×4 mm region.

phantom [9], scaled to 180 mm and sampled at 0.5 mm as partial-volume voxels ($282 \times 364 \times 324$). We consider both the original piecewise-constant phantom and a variant with smoothly varying interior intensity (approximately 10% brighter at the centre).

All models are fitted to the same voxel data. The polyhedral model assigns one intensity per region, while the tetrahedral model subdivides the regions into constant-intensity tetrahedra of at most 5 mm^3, with a coarser variant matched to the Gaussian parameter count. The Gaussian mixture contains 315 k primitives, fitted in two minutes (1500 iterations, 8192 points for TV estimation per iteration, $\lambda = 10^{-4}$). We report whole-volume image PSNR, model size, and k-space evaluation time and peak GPU memory for 32,768 samples on an RTX 6000 Ada.

As Table 1 and Fig. 3 show, the Gaussian mixture achieves the highest PSNR for both subjects, with the smallest storage, the fastest k-space evaluation and the lowest GPU memory.

Table 1. 3D Shepp–Logan benchmark. Whole-volume image PSNR for piecewise-constant (const.) and smoothly varying (smooth) intensities. Model size, evaluation time and peak GPU memory are reported for 32,768 k-space samples. Parameter counts include vertex coordinates and intensities for meshes, and 11 parameters per Gaussian.

	Param.	PSNR (dB)		Size	Time	Mem.
Model	(M)	const.	smooth	(MiB)	(s)	(GiB)
Polyhedral [10]	2.65	50.5	38.6	283	4.61	0.34
Tetrahedral	13.35	49.3	49.1	2559	37.5	2.51
Tetra., matched	3.42	39.7	39.9	703	10.4	0.71
Gaussian (ours)	3.47	**52.3**	**52.3**	**13**	**0.08**	**0.06**

3.2. Discretization error and reconstruction accuracy

We use a continuous BrainWeb reference fitted to eleven 0.5 mm tissue maps [13], comprising 1.70 M Gaussian primitives. Fitting

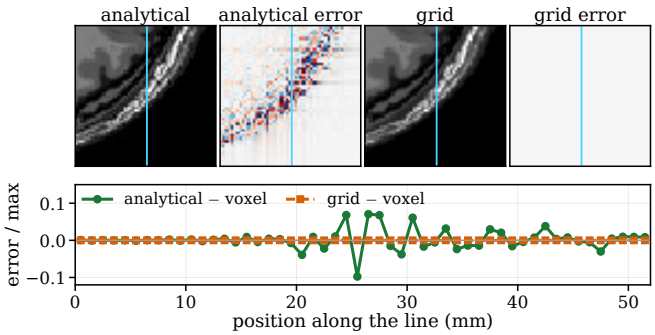


Fig. 4. Fully sampled 1 mm Cartesian acquisition of the BrainWeb subject without noise (posterior scalp and skull). Analytical data and the DFT of the 1 mm voxel phantom, both reconstructed by inverse DFT, with their errors against the voxel phantom ($\pm 10\%$ of the maximum). Bottom: both errors along the marked line.

takes 11 minutes on two GPUs, and tissue signals are computed using 1.5 T relaxation parameters. This reference is also used in the following motion experiments.

Spectral replication. Consider a voxel grid $\mathbf{r_n} = \mathbf{r}_0 + \Delta x\, \mathbf{n}$ storing either point samples $g = \rho$ or voxel averages $g = \rho * \Pi_{\Delta x}$, where $\Pi_{\Delta x}$ is a normalized box kernel [14]. By Poisson summation, its discrete Fourier sum is given by

$$\hat{y}_{\Delta x}(\mathbf{k}) = \sum_{\mathbf{q}\in\mathbb{Z}^3} \hat{g}\Big(\mathbf{k} + \frac{\mathbf{q}}{\Delta x}\Big) e^{i2\pi \mathbf{q}^\top \mathbf{r}_0/\Delta x}, \tag{9}$$

where $\hat{g} = \hat{\rho}$ for point samples and $\hat{g} = H_{\Delta x}\hat{\rho}$ for voxel averages, with $H_{\Delta x}(\mathbf{k}) = \prod_i \operatorname{sinc}(k_i \Delta x)$. For point samples, the $\mathbf{q} = \mathbf{0}$ term is the desired spectrum, while the remaining terms are spectral replicas. Using the Gaussian spectrum from Eq. (4), their contribution is bounded by

$$\begin{aligned} \big|\hat{y}_{\Delta x}(\mathbf{k}) - \hat{\rho}(\mathbf{k})\big| &\le \sum_n \big|S(\theta, \eta_{\tau(n)})\, m_n\big|\, \epsilon_n(\mathbf{k}), \\ \epsilon_n(\mathbf{k}) &= \sum_{\mathbf{q}\neq\mathbf{0}} e^{-2\pi^2 \sigma^2_{\min,n} \|\mathbf{k}+\mathbf{q}/\Delta x\|^2}, \end{aligned} \tag{10}$$

where $\sigma_{\min,n}$ is the smallest standard deviation of primitive n. The bound relates spectral replication to Gaussian width and grid spacing. Voxel averaging attenuates the replicas but also introduces in-band attenuation. We evaluate the resulting discretization error below.

Gibbs ringing. A real scanner acquires the continuous spectrum of the object only within a finite band, so its fully sampled reconstructions exhibit Gibbs ringing at sharp boundaries. Analytical simulation reproduces this effect: its data sample the continuous spectrum and are truncated at the same band edge, and their inverse DFT rings by up to about 10% of the image maximum at the scalp and skull (Fig. 4). DFT simulation performed on the reconstruction grid returns the discretized phantom without revealing this effect. Reconstructions evaluated against the grid from which their data were simulated are thus spared these errors, yielding over-optimistic results. Using a finer simulation grid can recover the ringing and reduce discretization errors, but requires additional spatial resolution. Analytical encoding instead evaluates the continuous object's spectrum directly, without this intermediate discretization.

Noise-independent accuracy ceiling. We simulate fully sampled 1 mm Cartesian and Nyquist-sampled 3D radial acquisitions

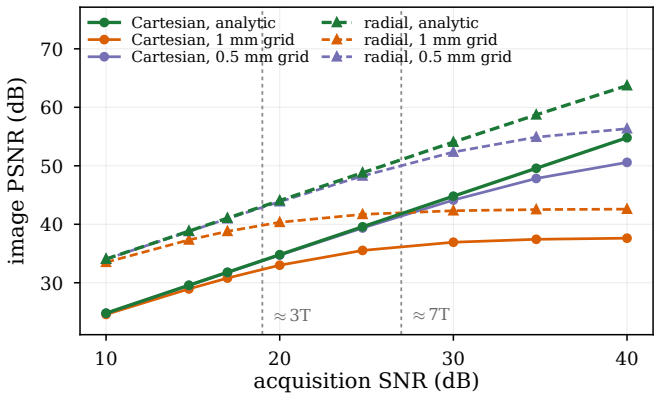


Fig. 5. Reconstruction accuracy versus acquisition SNR for fully sampled 1 mm Cartesian and 3D radial BrainWeb acquisitions. Analytical signals are compared with NUFFT simulations of 1 and 0.5 mm voxel-averaged representations under matched noise and reconstruction. Radial acquisitions use the same noise standard deviation as the Cartesian data. Dotted lines: scanner-calibrated reference SNRs for 1 mm multi-channel MPRAGE at 3 T and 7 T.

(16.3 M radial samples) of the T_1-weighted BrainWeb subject. Voxel-based signals are generated by NUFFT [15, 16] in single precision (tolerance 10^{-6}) from 1 and 0.5 mm rasters, whose voxel averages are estimated using a 0.25 mm sub-grid. Analytical and voxel-based data receive matched noise and the same reconstruction: inverse DFT for Cartesian data and, for radial data, 30 LSQR iterations (MRI-NUFFT) initialized with the density-compensated adjoint. Image PSNR is measured against the noiseless analytical reconstruction (Fig. 5).

At low SNR, reconstruction error is dominated by noise. As SNR increases, the voxel-based reconstructions reach PSNR ceilings of 37.7 dB (Cartesian) and 42.6 dB (radial) for the 1 mm raster, becoming apparent around acquisition SNRs of 23 and 18 dB, respectively. Refining the raster to 0.5 mm, at eight times the voxel count, raises these ceilings to 52.7 and 57.2 dB. In contrast, the analytical reconstructions continue to improve with SNR over the evaluated range. For reference, scanner-calibrated SNRs for 1 mm multi-channel MPRAGE acquisitions at 3 T and 7 T are approximately 19 and 27 dB under our definition, although these values depend on acquisition conditions. At these levels, the 1 mm grid incurs approximately 1.5–3 and 6–9 dB of PSNR loss, respectively. Finer grids reduce the mismatch at increased computational cost, whereas analytical encoding avoids the additional raster-discretization error for the fitted continuous object.

3.3. Multi-coil and motion simulation

Parallel imaging. We represent a 20-channel Biot–Savart array using 125 shared Fourier modes, the fitted maps define the coil sensitivities of the simulated phantom, for which multi-coil k-space is evaluated analytically [8]. For a 1 mm 3D radial acquisition with 361 k samples per channel (20-fold undersampling relative to the $182 \times 218 \times 182$ Cartesian acquisition, 7.2 M samples), multi-coil encoding takes 10.3 minutes and 0.34 GiB on an RTX 6000 Ada. Compared with 4.7 s for a single channel, the 131-fold increase is close to the 125 Fourier modes, reflecting the shared spectral evaluations across coils.

Rigid motion. We evaluate motion simulation on the BrainWeb subject using a T_1-weighted spoiled GRE model ($T_R/T_E = 15/3$ ms, $\alpha = 30°$). The 1 mm Cartesian acquisition ($216 \times 248 \times 216$, enlarged so that the moving head stays in the FOV) uses one readout per TR. Each A–P phase-encoding plane comprises 216 readouts (3.24 s) acquired at a fixed rigid pose, giving 248 poses over 13.4 min. The trajectory combines slow drift and abrupt displacements of up to 3.4 mm and 5°.

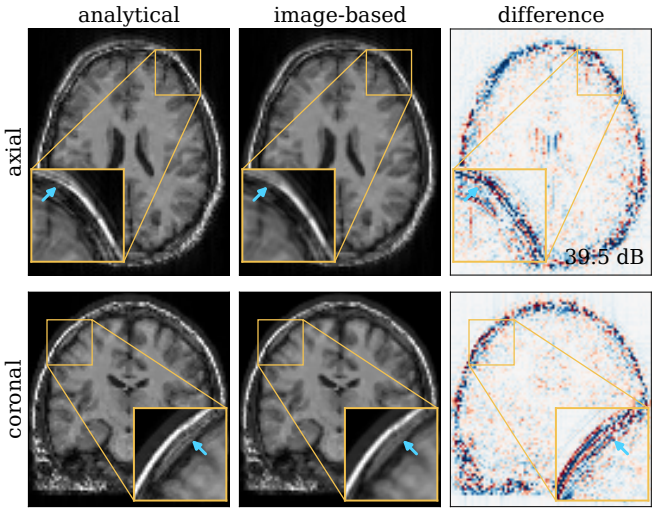


Fig. 6. Motion-corrupted BrainWeb acquisition with one rigid pose per phase-encoding plane (3.24 s), including displacements up to 3.4 mm and 5°. Analytical and image-based simulations are reconstructed by inverse DFT without motion correction; rows: axial and coronal slices of the same 3D volumes. The difference (image-based minus analytical, $\pm 8\%$ of the reference maximum) is the error added by image-based simulation. Insets: a 40 mm region near the scalp and skull; arrows: an edge blurred by resampling.

Analytical signals are evaluated directly at each pose using Eq. (8). For comparison, the image-based simulator applies trilinear resampling to a motion-free reconstructed volume at each pose, then Fourier-transforms the resampled image. Both methods assemble the acquired planes and reconstruct by inverse DFT without motion correction.

The image-based simulation differs from the analytical reference by 39.5 dB PSNR (SSIM 0.986; 39–40 dB across eight trajectories), with errors concentrated near anatomical boundaries (Fig. 6).

4. CONCLUSION

We presented an analytical MRI simulation framework based on anisotropic Gaussian mixtures fitted to anatomical volumes, supporting closed-form multi-coil encoding, sequence contrast and rigid motion. The representation achieved higher accuracy on spatially varying anatomy with substantially lower storage and evaluation cost than the tested polyhedral and tetrahedral models. Experiments showed that voxel-based simulation introduces resolution-dependent discretization errors, while image resampling adds errors under motion. Finer grids reduce these errors at increased computational cost, whereas analytical encoding evaluates the fitted continuous model directly without additional voxel discretization or motion resampling. The framework thus provides a practical continuous reference for MRI reconstruction studies.

5. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

6. ACKNOWLEDGMENTS

This work was supported by WP2 (Optimizing Workflows Efficiency) of the European Joint Virtual Lab on Artificial Intelligence, Data Analytics and Scalable Simulation (AIDAS), an initiative of FZJ and CEA. This work was performed using HPC resources from GENCI-IDRIS (Grant 2026-AD011017652). The authors have no relevant financial or non-financial interests to disclose.

AI tools were used to assist with implementation and to improve the clarity and presentation of the manuscript. All AI-assisted outputs were reviewed and supervised by the authors. The authors take full responsibility for the methods, results, interpretations, and content of the manuscript.

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